

SU TUNG LOC

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SKILLS

HPC:	Slurm, Spack, Warewulf, Apptainer, Docker, Linux, HPL and HPCG benchmarking.
DRL methods:	Reinforce, PPO, Q-learning, DQN, SARSA, A2C.
Machine Learning:	Python, PyTorch, TensorFlow, Scikit-learn, YOLO, RT-DETR, OpenCV, Gymnasium.
AI:	LangChain, LLMs, RAG, Prompt Engineering.
Parallel computing:	C, MPI, OpenMP, CUDA.
Cybersecurity & Networking:	Wireshark, Nmap, netcat, Burp Suite, Ghidra
SW/Web Development and GIS:	Java, JavaScript, Git, VS Code, SQL, NoSQL, HTML, CSS, Bootstrap, Leaflet, D3, DC.js
Others:	AWS, Weights & Biases (wandb), Anyscale, Foxglove, RViz2, Glim.
Languages	English, Vietnamese, Chinese

EDUCATION

Master of Science in Computer Science	Expected: May 2026
<i>University of New Mexico</i>	GPA: 4.18/4.0
Bachelor of Science in Computer Science	May 2024
<i>University of New Mexico</i>	<i>Magna Cum Laude</i> GPA: 3.88/4.0
Associate Degree	Aug 2019
<i>Peninsula College</i>	<i>Magna Cum Laude</i> GPA: 3.92/4.0

PROJECTS

HPL benchmarking | *Slurm, Bash, Git, OpenMPI, OpenBLAS*

- Performed HPL benchmarking on a multi-node HPC cluster using Slurm, OpenMPI, optimizing build parameters along with MPI communication to measure peak FLOPS performance, and evaluate the cluster architecture behavior.

HPCG benchmarking | *Slurm, Apptainer, Git, OpenMPI, NVIDIA container*

- Performed HPCG benchmarking on both CPU and GPU configurations using MPI and Slurm to evaluate memory bandwidth, communication overhead, and real-world solver performance.

Lunar Landar Training Analysis | *Python, PyTorch, OpenAI, Gymnasium, Matplotlib*

- Implemented and compared REINFORCE, SARSA, and DQN algorithms for Lunar Lander using PyTorch and Gymnasium, analyzing how exploration strategies and hyper-parameters impact learning performance.

Reinforcement Learning for Motion Planning and the Control of Dynamical Systems | *Python, PyTorch*

- Developed a 2D motion planning and vehicle control system using Q-learning, creating a custom RL environment and agent for automatic parking and optimal path navigation based on ODE-based simulations.

Driving Assistant | *Python, PyTorch, OpenCV, HuggingFace, YOLO*

- Built a Python-based dash-cam analysis system that integrates object detection, LLM-generated commentary, and automated pipelines to detect road objects, produce annotated video outputs, and generate contextual driving alerts.

Implement PPO on Atari Pong | *Python, PyTorch, OpenCV, Numpy, OpenAI, Gymnasium, Matplotlib*

- A state-of-the-art deep reinforcement learning algorithm that train an agent in playing Pong game

NYC Tree Census Web Map | *HTML/CSS/JavaScript, Leaflet, DC, D3, Crossfilter, Bootstrap*

- Developed an interactive web map visualizing the 2015 NYC Tree Census, enabling exploration of tree counts, species, and health status across all five boroughs through dynamic charts and filters.

Hexapawn | *Java, JavaFX*

- Developed a Hexapawn game in Java utilizing heuristic learning approach to create an adaptive unbeatable computer opponent.

Modulo time table visualization | *Java, JavaFX*

- Developed an interactive data visualization tool to map modular arithmetic multiplication tables, translating abstract number theory into dynamic geometric patterns.

Live Auction Distribution | *Java, JavaFX, Socket, Network*

- Developed an interactive web map visualizing the 2015 NYC Tree Census, enabling exploration of tree counts, species, and health status across all five boroughs through dynamic charts and filters.

Research

Countering of Block Withholding for Miners |

- Developed a theoretical model and simulation to analyze Bitcoin Block Withholding attacks, demonstrating that honest miners can protect their profitability by optimally allocating computational resources between pool and solo mining.

Analysis of Distributed Denial of Service Attacks: An Internal Perspective |

- Conducted a theoretical and empirical analysis of Distributed Denial of Service attacks by simulating attack scenarios using Docker containers and testing against a live Cloudflare-protected website, demonstrating that high concurrency requests exponentially degrade server performance beyond a critical threshold

Analysis of Censorship Evasion Tools |

- Evaluated the performance, privacy, and obfuscation capabilities of NordVPN, Tor, and Psiphon through empirical testing and case studies across high-censorship regions.

EXPERIENCE

AI Research Assistant May 2026 - Present

UNM x Multiply Lab
* Conducted researches to develop robotic system that enable the manufacturing of advanced next-generation therapies at scale and affordable.

Teaching Assistant Aug - June 2026

University of New Mexico
* Assisted faculty with course instruction, project guidance, and development of teaching materials.
* Held student consultations for homework and graded exams, quizzes, assignments, and papers.

TechDen Sales Assistant June - Aug 2025

University of New Mexico
* Managed and updated product data using internal databases and spreadsheets, ensuring accurate inventory records
* Assisted with website updates, including product listings and content changes, to maintain current and accurate information.

Grader CS105/CS341L Aug 2024 - May 2025

University of New Mexico
* Implemented new assignments. Reviewed and graded homework assignments according to established standards, and assisted students with questions to clarify concepts.

Student Engineering Intern May - Aug 2024

Explora x UNM CHTM
* Assisted in projects involving advanced laser and microwave technologies while supporting STEM outreach initiatives for future generations.
* Performed clean room operations and conducted experiments using Electron Microscopy, Diode Analysis, XRD, and Photoluminescence spectroscopy.